MEDIUM · ALGEBRA

Algebra

02

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 GRID-IN ALGEBRA

What is the slope of the graph of $y = \frac{1}{4}27x + 15 + 7x$ in the xy-plane?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q2

GRID-IN ALGEBRA

If $2(3t - 10) + t = 40 + 4t$, what is the value of $3t$?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q3 ALGEBRA

In the linear function f, $f0 = 8$ and $f1 = 12$. Which equation defines f?

A. $fx = 12x + 8$

B. $fx = 4x$

C. $fx = 4x + 12$

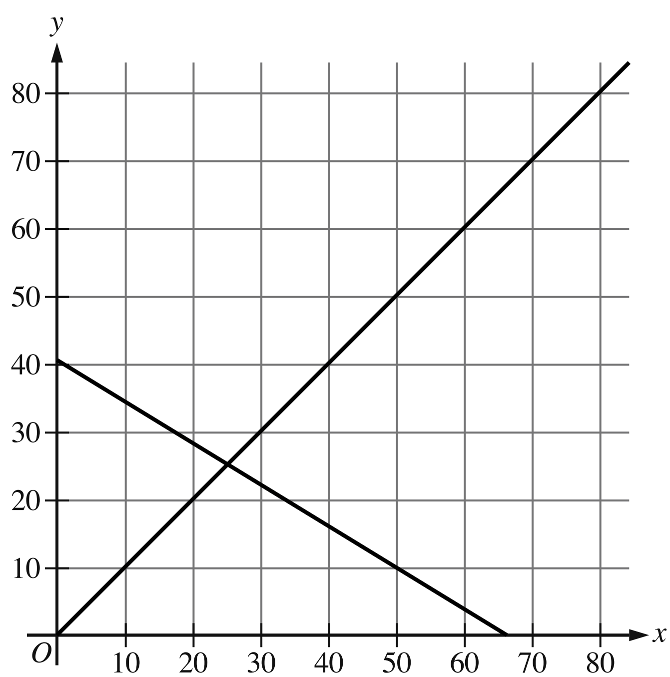
D. $fx = 4x + 8$

$$x + y = 53$$

$$11x + 18y = 730$$

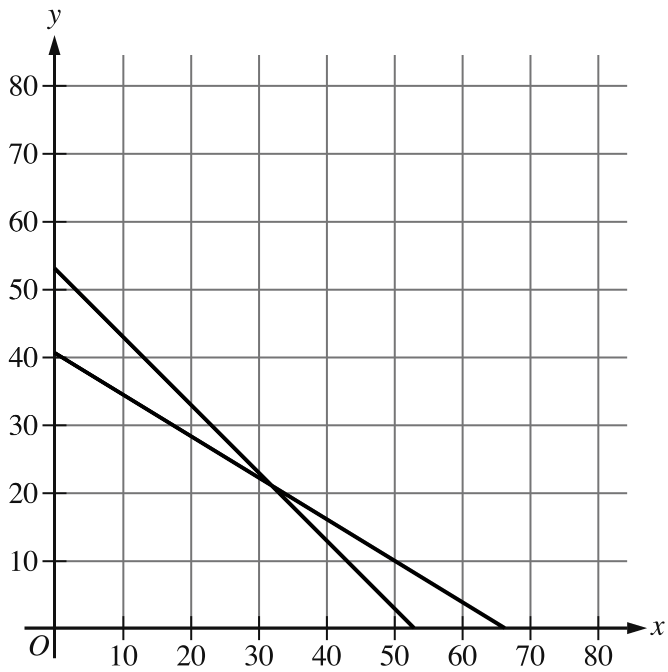
The given equations represent the possible numbers of beach chairs, x, and umbrellas, y, rented at a park last month and the total spent, in dollars, to rent those beach chairs and umbrellas. Which of the following graphs represents this situation?

A.



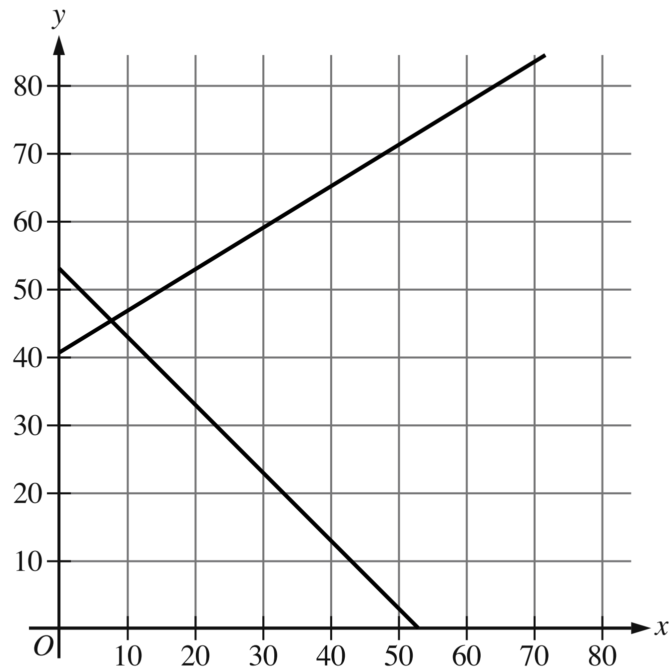
- For the first line in the system:
  - The line slants gradually down from left to right.
  - The line passes through the following approximate points:
    - (0 comma 41)
    - (66 comma 0)
- For the second line in the system:
  - The line slants gradually up from left to right.
  - The line passes through the following points:
    - (0 comma 0)
    - (50 comma 50)
- The 2 lines intersect at 1 point.

B.



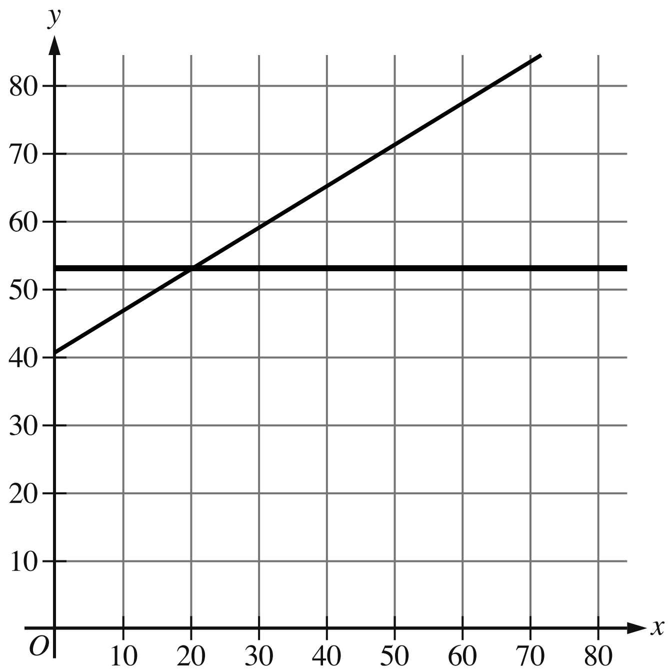
- For the first line in the system:
  - The line slants gradually down from left to right.
  - The line passes through the following points:
    - (0 comma 53)
    - (53 comma 0)
- For the second line in the system:
  - The line slants gradually down from left to right.
  - The line passes through the following approximate points:
    - (0 comma 41)
    - (66 comma 0)
- The 2 lines intersect at 1 point.

C.

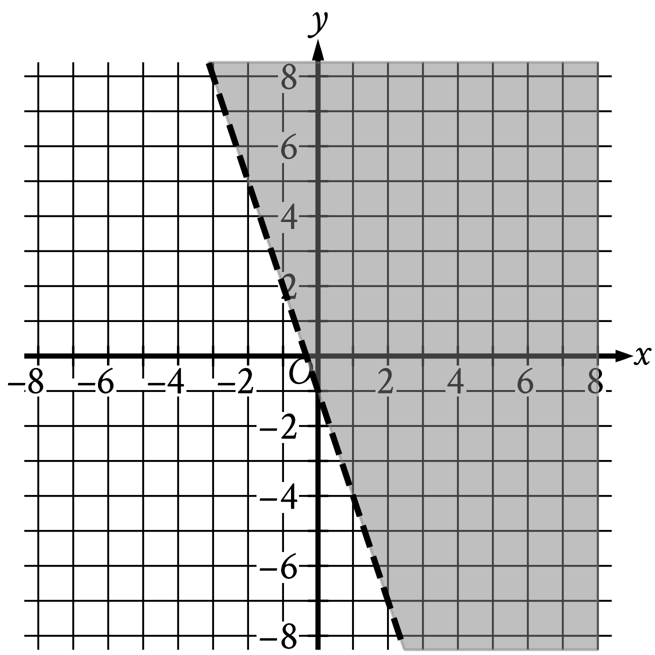


- For the first line in the system:
  - The line slants gradually down from left to right.
  - The line passes through the following points:
    - (0 comma 53)
    - (53 comma 0)
- For the second line in the system:
  - The line slants gradually up from left to right.
  - The line passes through the following approximate points:
    - (0 comma 41)
    - (30 comma 59)
- The 2 lines intersect at 1 point.

D.



- For the first line in the system:
    - The line is horizontal.
    - The line passes through the following points:
        - (0 comma 53)
        - (30 comma 53)
- For the second line in the system:
    - The line slants gradually up from left to right.
    - The line passes through the following approximate points:
        - (0 comma 41)
        - (30 comma 59)
- The 2 lines intersect at 1 point.



- The boundary of the inequality is a dashed line.
  - The line slants sharply down from left to right.
  - The line passes through the following points:
    - (negative 1 comma 2)
    - (0 comma negative 1)
- The area above and to the right of the boundary is shaded.

The shaded region shown represents the solutions to which inequality?

A. $y < -1 + 3x$

B. $y < -1 - 3x$

C. $y > -1 + 3x$

D. $y > -1 - 3x$

Q6 ALGEBRA

What is the slope of the graph of $10x - 5y = -12$ in the xy-plane?

A. -2

B. $-\frac{5}{6}$

C. $\frac{5}{6}$

D. 2

Q7 ALGEBRA

$$\frac{1}{4}x + 5 - \frac{1}{3}x + 5 = -7$$

What value of x is the solution to the given equation?

A. -12

B. -5

C. 79

D. 204

Q8

GRID-IN ALGEBRA

$$h(x) = x + b$$

For the linear function h, b is a constant and $h(0) = 45$. What is the value of b?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q9

ALGEBRA

$$y = 3x$$

$$2x + y = 12$$

The solution to the given system of equations is x, y. What is the value of $5x$?

A. 24

B. 15

C. 12

D. 5

Q10

ALGEBRA

A particular botanist classifies a species of plant as tall if its typical height when fully grown is more than 100 centimeters. Each of the following inequalities represents the possible heights h, in centimeters, for a specific plant species when fully grown. Which inequality represents the possible heights h, in centimeters, for a tall plant species?

A. $106 < h < 158$

B. $80 < h < 100$

C. $42 < h < 87$

D. $17 < h < 85$